

DATA PREPARATION AND SIMILARITY AND DISTANCES

DV2627 – Advanced Machine Learning

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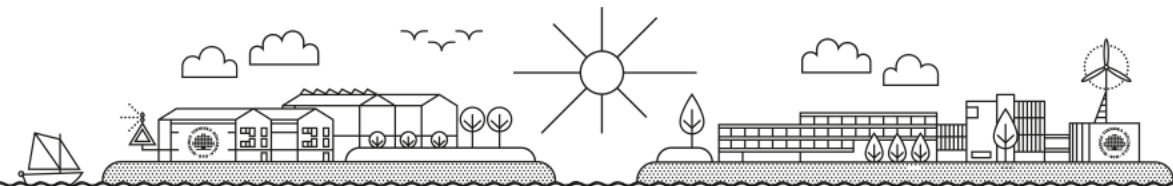
January 23, 2025



OUTLINE



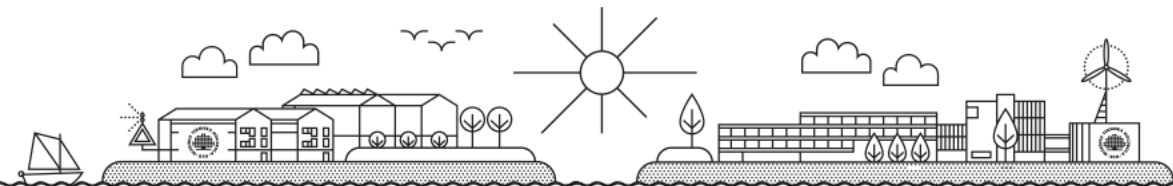
- Introduction
- Feature Extraction and Portability
- Data Cleaning
- Data Reduction
 - Sampling
 - Feature Subset Selection
- Similarity and Distances
 - Quantitative Data
 - Categorical Data
 - Temporal Similarity Measures
 - Continuous time series
 - Discrete sequences



INTRODUCTION

“Success depends upon previous preparation, and without such preparation there is sure to be failure.” —Confucius

- Feature Extraction and Portability
 - Raw logs, documents, semi-structured data
 - Data may contain heterogeneous types
- Data Cleaning
 - Missing, Erroneous, and Inconsistent
- Data Reduction, Selection, and Transformation
 - Efficiency, Effectiveness

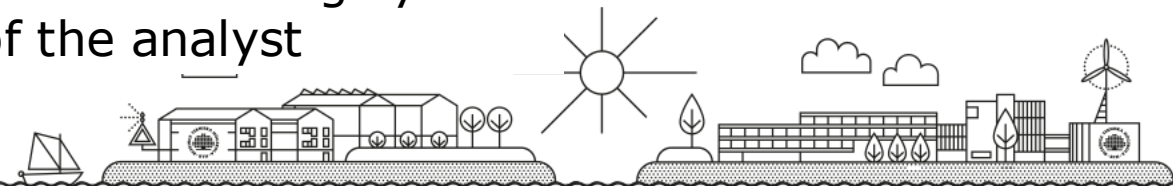


FEATURE EXTRACTION



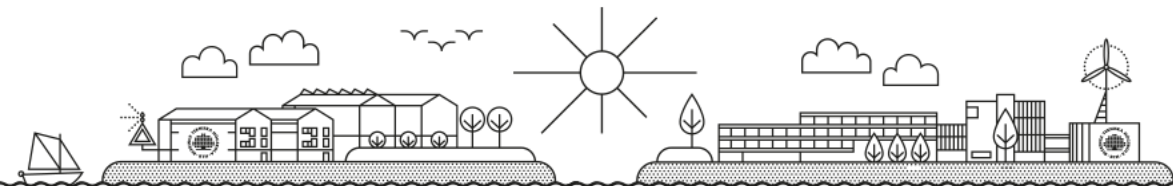
Domain	Raw Data (low level)	Features (high level)
Sensor	Low-level signals	Wavelet or Fourier transforms
Image	Pixels	Color histograms
Web logs	Text strings	IP address Action
Network traffic	Characteristics of the network packets	Number of bytes transferred Network protocol
Document data	Text strings	Bag-of-words Entity extraction

Feature extraction is an art form that is highly dependent on the skill of the analyst



DATA TYPE PORTABILITY (1)

- Data is often heterogeneous
 - A demographic data set may contain both numeric and mixed attributes
- Possible solutions
 - Designing an algorithm with an arbitrary combination of data types
 - Time-consuming and sometimes impractical
 - Converting between various data types
 - Utilize off-the-shelf tools for processing

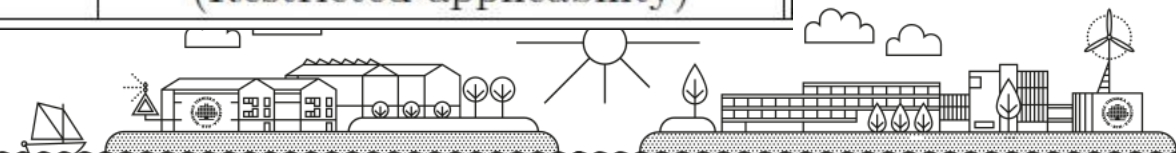


DATA TYPE PORTABILITY (2)

○ Ways of Transforming Data

Table 2.1: Portability of different data types

Source data type	Destination data type	Methods
Numeric	Categorical	Discretization
Categorical	Numeric	Binarization
Text	Numeric	Latent semantic analysis (<i>LSA</i>)
Time series	Discrete sequence	<i>SAX</i>
Time series	Numeric multidimensional	<i>DWT, DFT</i>
Discrete sequence	Numeric multidimensional	<i>DWT, DFT</i>
Spatial	Numeric multidimensional	2-d <i>DWT</i>
Graphs	Numeric multidimensional	<i>MDS</i> , spectral
Any type	Graphs	Similarity graph (Restricted applicability)

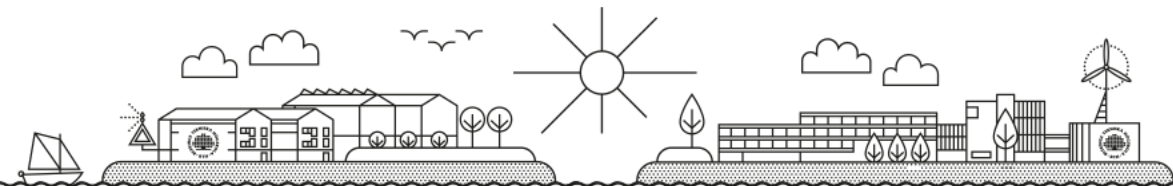


NUMERIC TO CATEGORICAL DATA: DISCRETIZATION (1)

- Divides ranges of the numeric attribute into Φ ranges

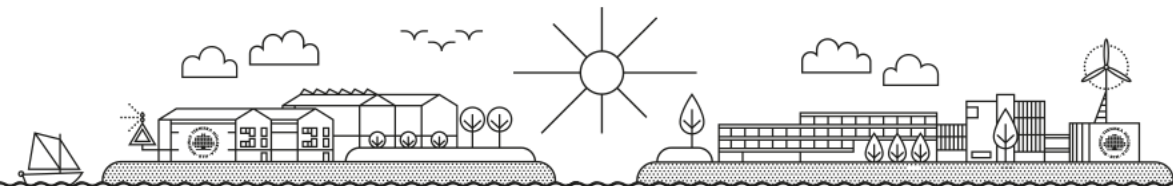


- Age Attribute
 - [0, 10], [11, 20], [21, 30] , ...
 - Symbolic value of any record in [11, 20] is “2”.
- Salary
 - [40,000, 80,000], [1,040,000, 1,080,000]



NUMERIC TO CATEGORICAL DATA: DISCRETIZATION (2)

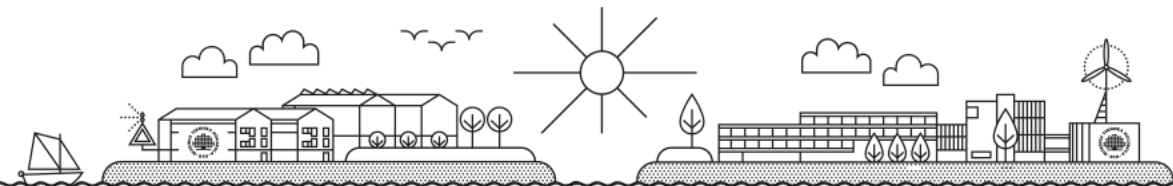
- Equi-width Ranges
 - Each range $[a, b]$ is chosen such that $b - a$ is a constant
- Equi-log Ranges
 - Each range $[a, b]$ is chosen such that $\log b - \log a$ is a constant
 - For example, $[a, a \cdot \alpha]$, $[a \cdot \alpha, a \cdot \alpha^2]$, $[a \cdot \alpha^2, a \cdot \alpha^3]$, ... where $\alpha > 1$
- Equi-depth Ranges
 - Each range has an equal number of records
 - Sorting and Selecting



CATEGORICAL TO NUMERIC DATA: BINARIZATION

- Two categories
 - 0,1 or $-1,1$
- Φ categories
 - Φ -dimensional indicator vector
 - The position of 1 indicates the category

$$\begin{array}{c} \text{R} \quad \text{G} \quad \text{B} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \Phi = 3 \end{array}$$



TEXT TO NUMERIC DATA

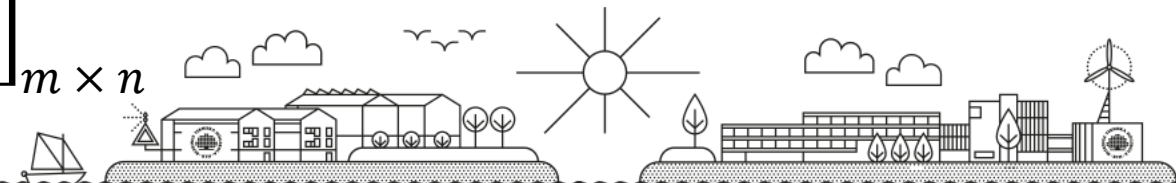
- Tokenization, Stop Word Removal, Stemming, Weighting (TF-IDF)
- Document-Term Matrix

	this	example	...
This is a sample.	1	0	...
This is another example.	1	1	...
...	...	...	...

 $m \times n$

- Term Frequency–Inverse Document Frequency (TF-IDF)

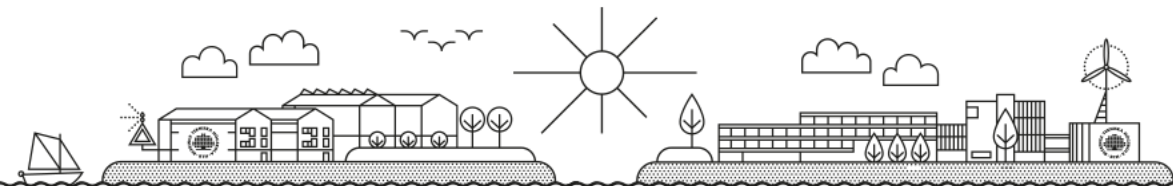
	this	example	...
This is a sample.	0	0	...
This is another example.	0	0.129	...
...	...	...	...

 $m \times n$


WORD WEIGHTING: TF-IDF

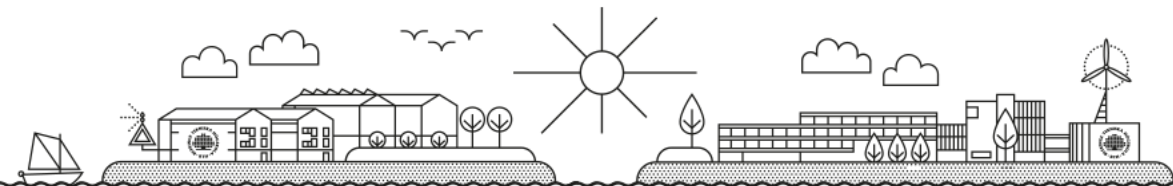
- $tf-idf(t, d, D) = tf(t, d) \cdot idf(t, D)$
 - t stands for a term (word)
 - d stands for a document
 - D stands for a document set
- $tf(t, d) = \frac{\text{count. of term } t \text{ in a document}}{\text{no. of terms in the document}}$
- $idf(t, D) = \log_{10} \left(\frac{\text{no. of documents}}{\text{no. of documents contain } t} \right),$

where 0 means the term is common while 1 means it is rare

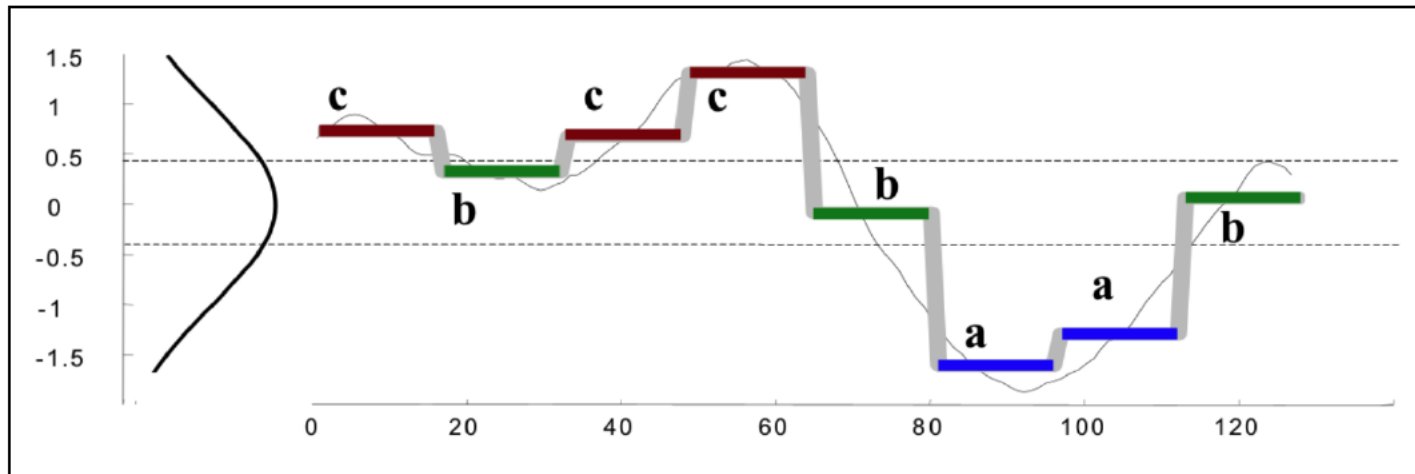


TIME SERIES TO DISCRETE SEQUENCE DATA

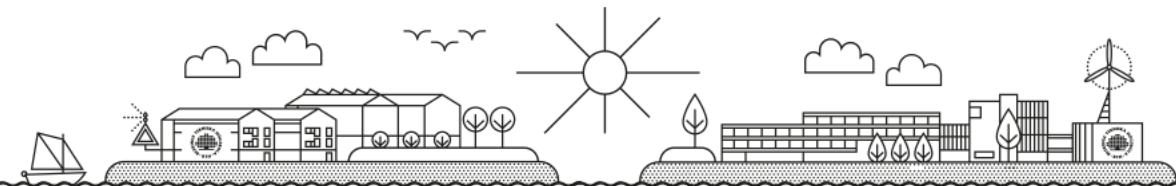
- Symbolic Aggregate Approximation (SAX)
 - 1. Window-based averaging
 - Divide the time series into the window of size w
 - Compute the average value in each window
 - 2. Value-based discretization
 - Discretize the average value by equi-depth intervals
 - ✓ Each symbol has an approximately equal frequency
- How to Ensure Equi-depth?
 - Assume certain distribution, such as Gaussian
 - Boundaries of the intervals are based on **quantiles** of the Gaussian distribution



TIME SERIES TO DISCRETE SEQUENCE DATA

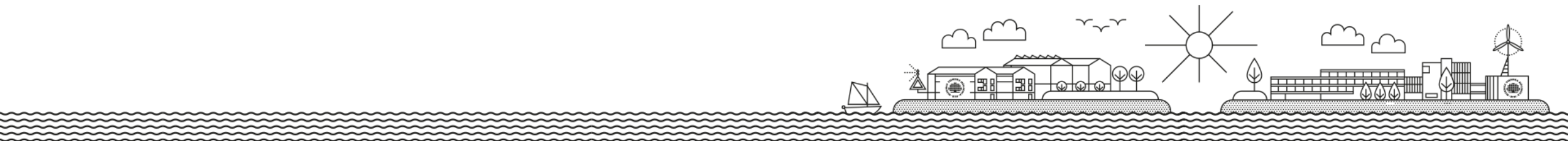


Lin J, Keogh E, Lonardi S, Chiu B. A symbolic representation of time series, with implications for streaming algorithms. In Proceedings of the 8th ACM SIGMOD workshop on Research issues in data mining and knowledge discovery 2003 Jun 13 (pp. 2-11).



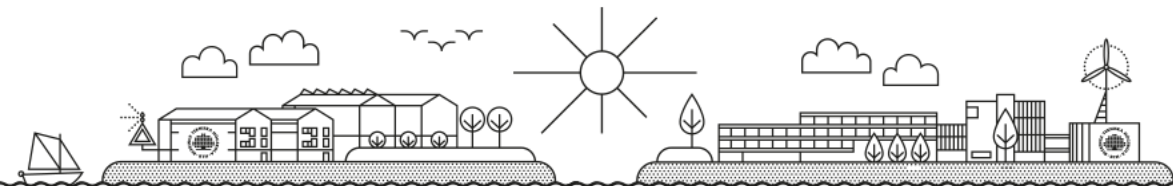
THE REASON OF CLEANING

- Data collection technologies are inaccurate
 - Sensors – hardware limitations (collection/transmission)
- Privacy reasons
- Manual errors
- Data collection is expensive
 - Medical test



HANDLING MISSING ENTRIES

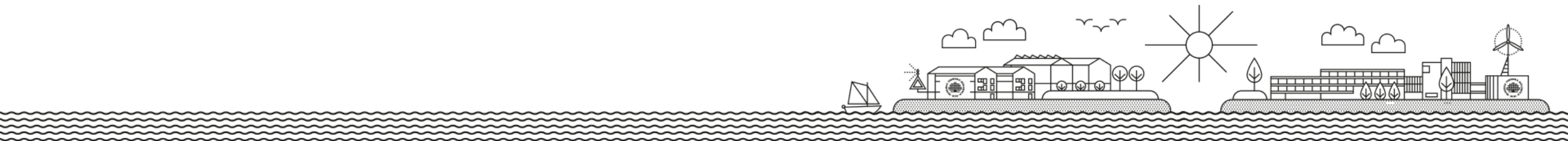
- Delete the data record contains missing entries
 - What to do if nothing left?
- Estimate or Impute the Missing Values
 - Additional errors may be introduced
- Designing an algorithm that works with missing data
 - it avoids the additional biases inherent in the imputation process.



HOW TO HANDLE NOISY DATA?

○ Binning

- first sort data and partition into (equal-frequency) bins
- then one can smooth by bin's **mean**, **median**, **boundaries**, etc.



HOW TO HANDLE NOISY DATA?

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

○ Binning

- first sort
- then one

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15

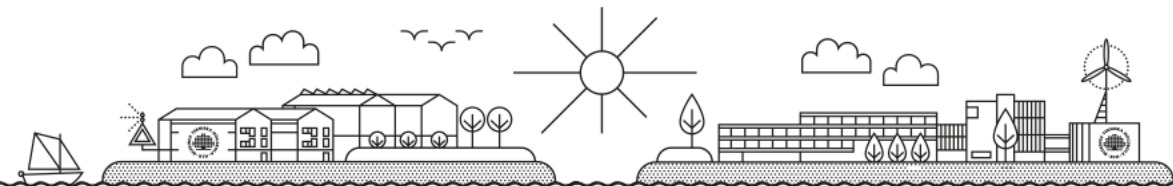
Bin 2: 21, 21, 24

Bin 3: 25, 25, 34



HOW TO HANDLE NOISY DATA?

- Binning
 - first sort data and partition into (equal-frequency) bins
 - then one can smooth by bin's **means**, **median**, **boundaries**, etc.
- Regression
 - smooth by fitting the data into regression functions
- Clustering
 - detect and remove outliers
- Combined computer and human inspection
 - detect suspicious values and check by humans (e.g., deal with possible outliers)



SCALING AND NORMALIZATION

- Features have different scales
 - Age versus Salary

- Standardization

- If the j th attribute has mean μ_j and standard derivation σ_j

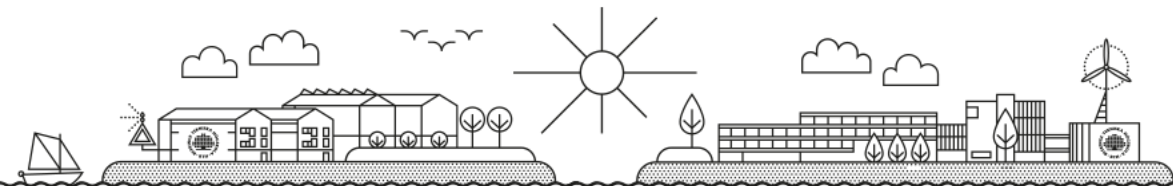
$$z_i^j = \frac{x_i^j - \mu_j}{\sigma_j}$$

- Majority of the normalized values will be in the range $[-3,3]$ under the **normal distribution assumption**.

- Min-Max scaling

- Map to $[0,1]$
 - Sensitive to noise and outliers

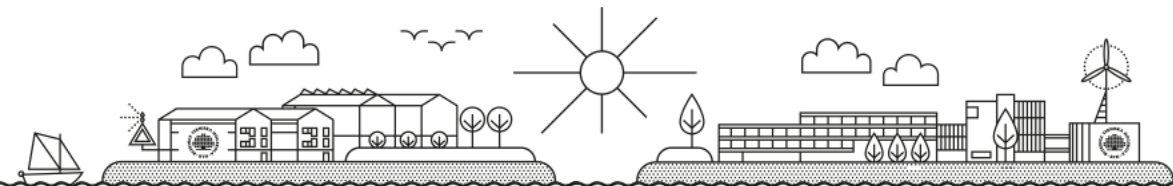
$$y_i^j = \frac{x_i^j - \min_j}{\max_j - \min_j}$$



DATA REDUCTION

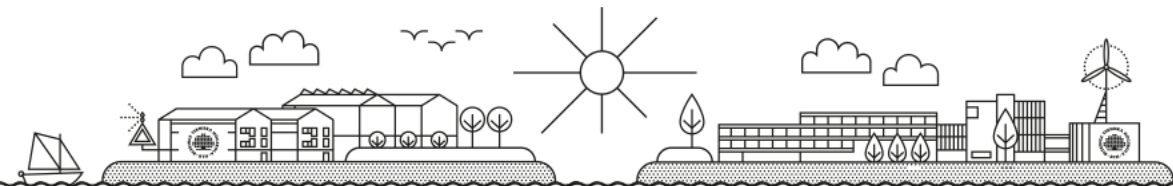
- The Advantages
 - Reduce space complexity
 - Reduce time complexity
 - Reduce noise
 - Reveal hidden structures

- The Disadvantages
 - Information loss



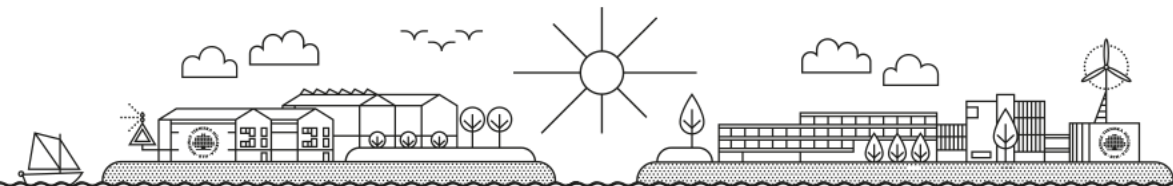
SAMPLING FOR STATIC DATA

- Unbiased (Uniform) Sampling
 - Sampling without replacement
 - Sampling with replacement
 - Duplicates are possible
- Biased Sampling
 - Some parts of the data are emphasized
- Stratified Sampling
 - Partition data into a set of strata (groups)
 - Sample in each of stratum



AN EXAMPLE OF SAMPLING

- There are 10,000 people which contain 100 millionaires
- Unbiased Sampling 100 people
 - In expectation, one millionaire will be sampled
 - In practice, maybe no millionaires are sampled
- Stratified Sampling
 - Unbiased Sampling 1 from 100 millionaires
 - Unbiased Sampling 99 from remaining



RESERVOIR SAMPLING FOR DATA STREAMS

- The Setting
 - Data arrive sequentially
 - We want sample k of them uniformly
 - There is a reservoir that can hold k data points
- The Algorithm (k sample size, n length of data stream)
 - The first k data points are kept
 - Insert the n th data point with probability k/n
 - If the n th data is inserted, then drop one of the existing k data points uniformly

Stream = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

1. Reservoir = [1, 2, 3, 4, 5], $k = 5$
2. Sampled data = [6, 2, 11, 8, 12]

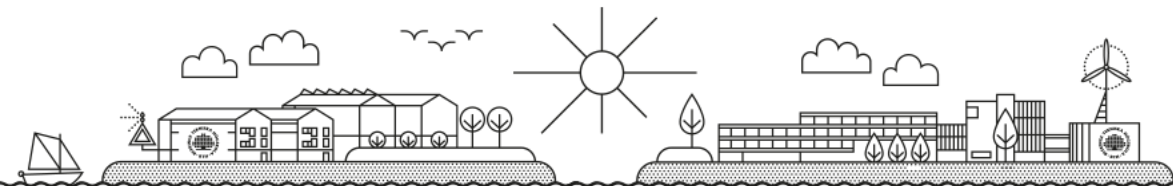
```
# A function to randomly select
# k items from stream[0..n-1].
def selectKItems(stream, n, k):
    i=0;
    # index for elements
    # in stream[]
    # reservoir[] is the output
    # array. Initialize it with
    # first k elements from stream[]
    reservoir = [0]*k;
    for i in range(k):
        reservoir[i] = stream[i];
    # Iterate from the (k+1)th
    # element to nth element
    while(i < n):
        # Pick a random index
        # from 0 to i.
        j = random.randrange(i+1);
        # If the randomly picked
        # index is smaller than k,
        # then replace the element
        # present at the index
        # with new element from stream
        if(j < k):
            reservoir[j] = stream[i];
        i+=1;
```

<https://www.geeksforgeeks.org/reservoir-sampling/>



FEATURE SUBSET SELECTION

- Unsupervised Feature Selection
 - Using the performance of **unsupervised** learning to guide the selection
 - E.g., Drop features with zero-variance
- Supervised Feature Selection
 - Using the performance of **supervised** learning to guide the selection
 - E.g., Wrapper feature selection: uses a model to score different subsets of features to finally select the best one.
 - Backward selection, Forward selection, Recursive Feature Elimination
- Check Feature selection module in scikit-learn



SIMILARITY AND DISTANCES

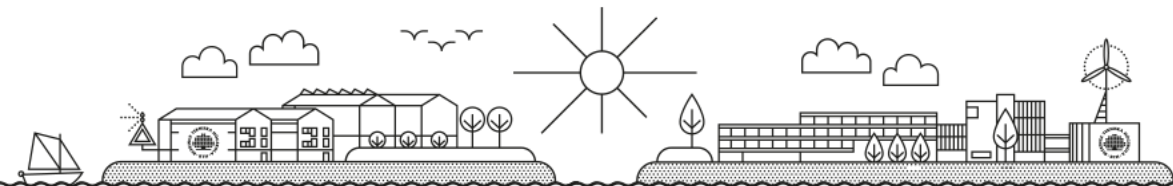
○ Definition

Given two objects O_1 and O_2 , determine a value of the similarity $Sim(O_1, O_2)$ (or distance $Dist(O_1, O_2)$) between the two objects.

- Distance functions for spatial data
- Similarity functions for text

○ Representation

- Closed-form, such as Euclidean distance
- Defined algorithmically



QUANTITATIVE DATA

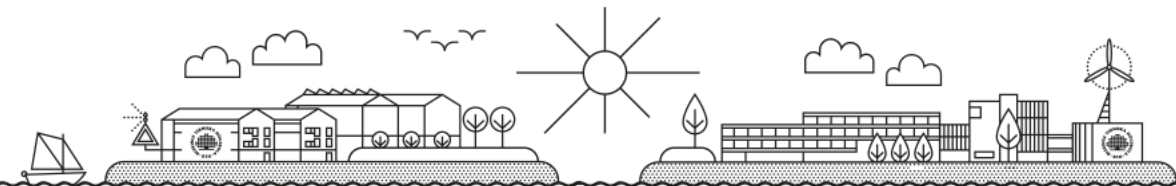
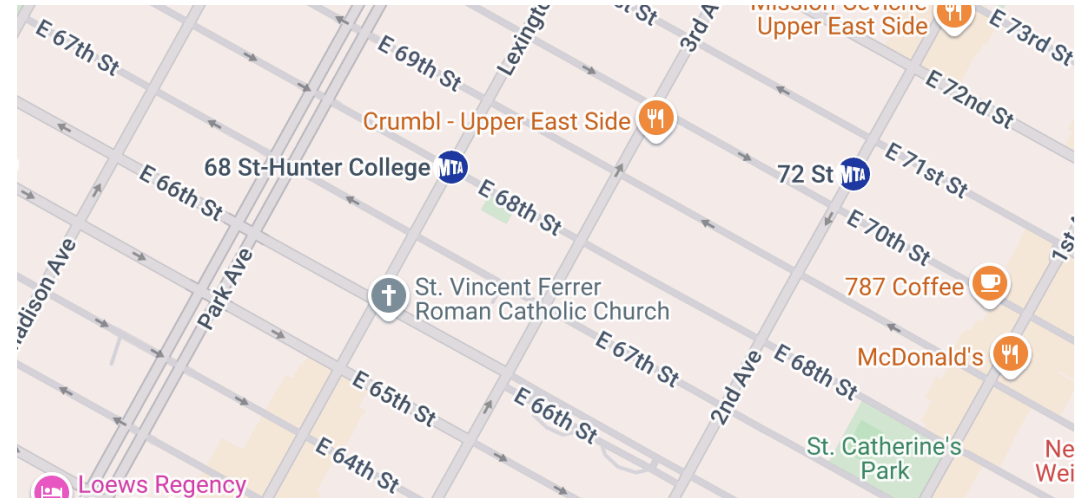


○ L_p -norm

$$\bar{X} = (x_1 \dots x_d) \text{ and } \bar{Y} = (y_1 \dots y_d)$$

$$Dist(\bar{X}, \bar{Y}) = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$$

- $p = 1$: Manhattan norm
 - Sum of absolute values
- $p = 2$: Euclidean norm
 - Square root of sum of squares
 - Rotation-invariant
- $p = \infty$: Infinity norm
 - Largest absolute value
 - Select the dimension when two objects are the most distant (all other features will be ignored).

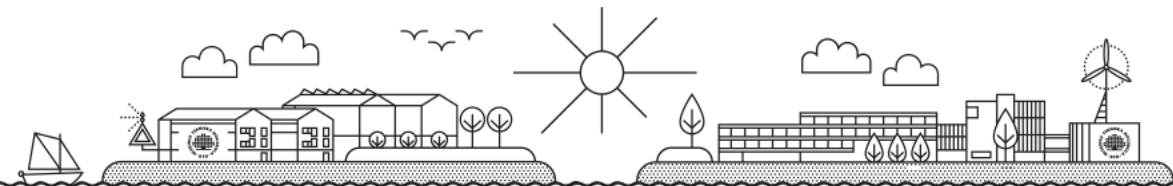


DOMAIN-SPECIFIC RELEVANCE

- Some Features are more Important
 - Credit-scoring
 - Salary is more important than Gender
- Generalized L_p - norm

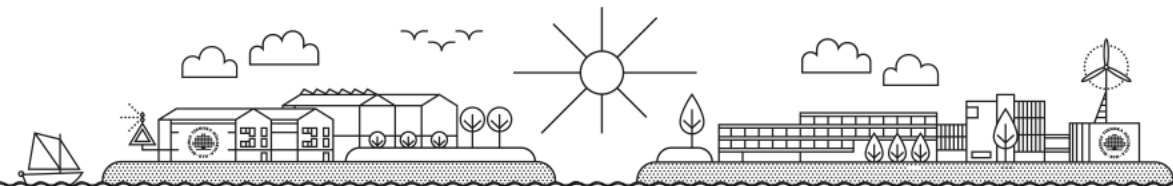
$$Dist(\overline{X}, \overline{Y}) = \left(\sum_{i=1}^d a_i \cdot |x_i - y_i|^p \right)^{1/p}$$

- $a_1, \dots, a_d$ are nonnegative coefficients
- Generalized Minkowski distance



ISSUES

- Curse of Dimensionality
 - Distance-based algorithms lose their effectiveness as the dimensionality increases
- Many features are likely to be irrelevant
 - Especially in high-dimensional data
- An Example
 - A cluster containing diabetic patients vs. epileptic patients
 - Blood glucose levels are more important for diabetes
 - while a different set of features is more important for epilepsy
- L_p - norms
 - Suffer from the additive noise effects of the irrelevant features
 - E.g, Euclidean metric may unnecessarily contribute a high value from the more noisy components because of its **square-sum** approach.
 - Do not behave in a similar way
 - When the dimensionality is high
 - When there exist irrelevant features

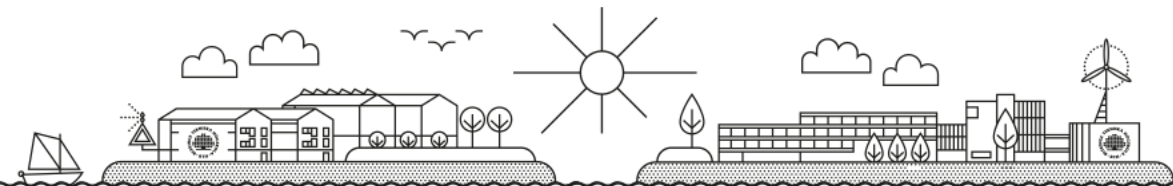


MATCH-BASED SIMILARITY COMPUTATION

- The key idea
 - De-emphasize irrelevant features
- Proximity Thresholding
 - Discretized each feature into k_d equidepth buckets
- Similarity Evaluation

$$\begin{aligned}\bar{X} &= [x_1, x_2, \dots, x_d] \longrightarrow [1, 3, \dots, k_d] \\ \bar{Y} &= [y_1, y_2, \dots, y_d] \longrightarrow [5, 3, \dots, k_d]\end{aligned}$$

- for dimension i , if both x_i and y_i belong to the same bucket, the two records are said to be in proximity on dimension i .
- $S(\bar{X}, \bar{Y}, k_d)$ is the set of features mapped to the same bucket

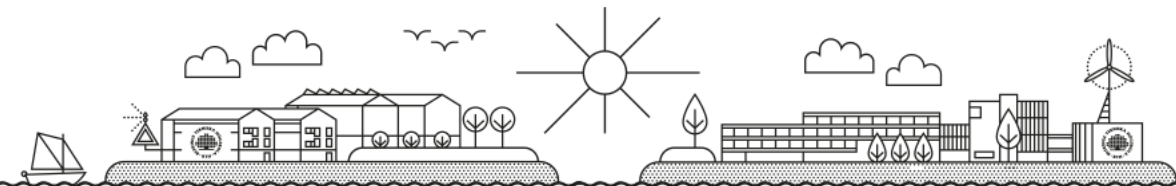


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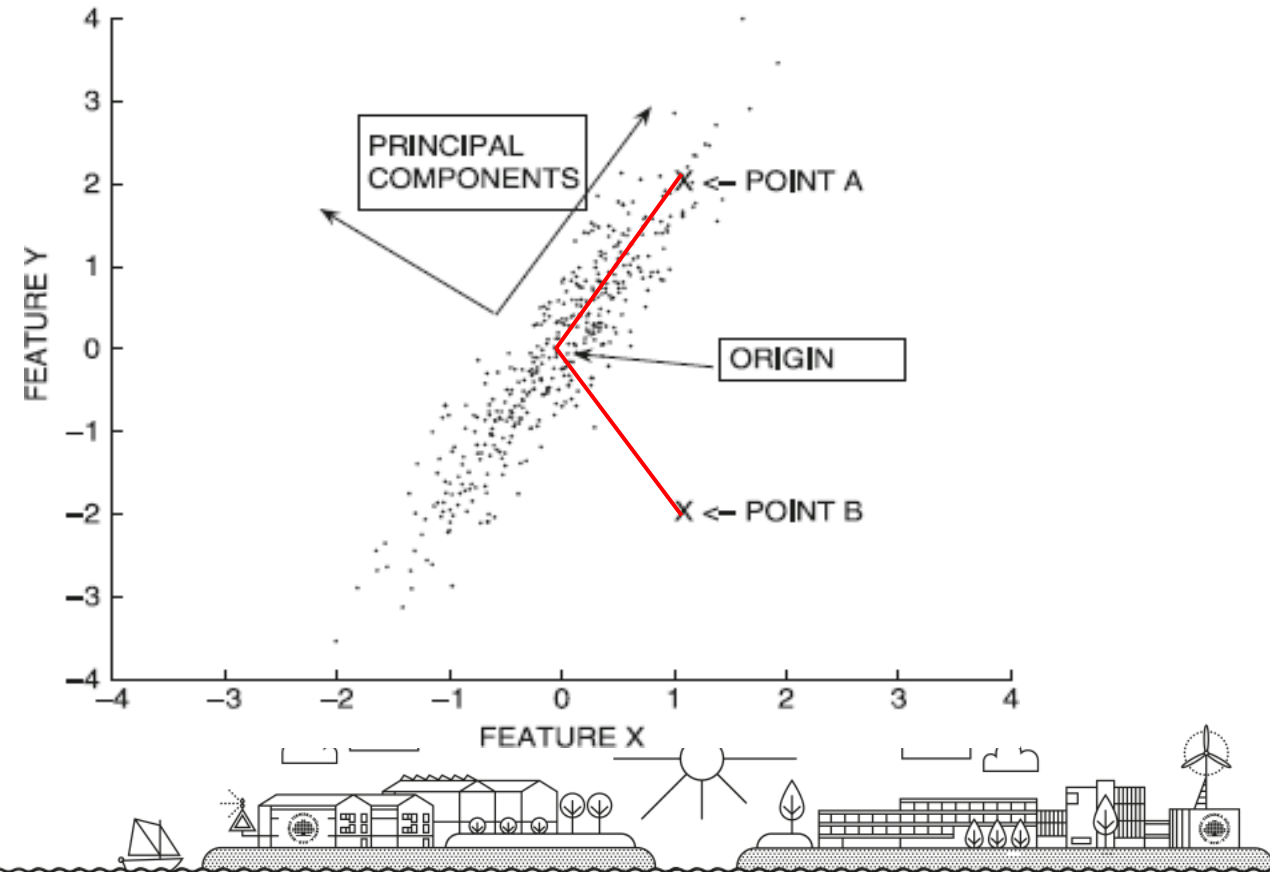
$$PSelect(\bar{X}, \bar{Y}, k_d) = \left[\sum_{i \in S(\bar{X}, \bar{Y}, k_d)} \left(1 - \frac{|x_i - y_i|}{m_i - n_i} \right)^p \right]^{1/p}$$

- for dimension i , if both x_i and y_i belong to the same bucket, the two records are said to be in proximity on dimension i .
- $S(\bar{X}, \bar{Y}, k_d)$ is the set of features mapped to the same bucket
- m_i and n_i are the upper and lower bounds of the bucket in dimension i .
- values 0 and $|S(\bar{X}, \bar{Y}, k_d)|$ where larger values mean higher similarity



DATA DISTRIBUTION (1)

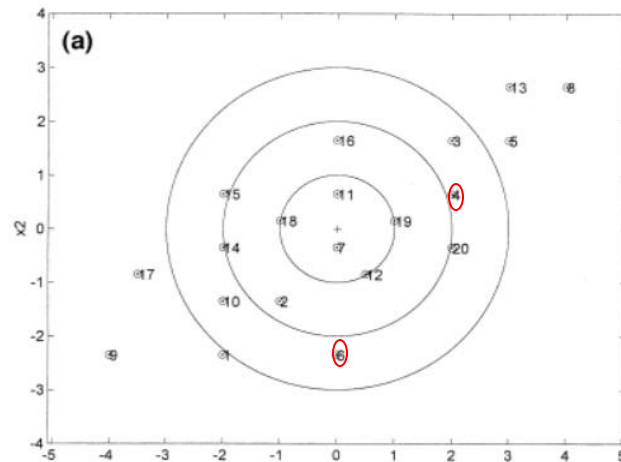
- $A = (1,2)$, $B = (1,-2)$, and $O = (0,0)$
 - O-A is aligned with a **high-variance** direction.
 - O-B is aligned with a **low-variance** direction.
 - O-A is more likely to be less than O-B.



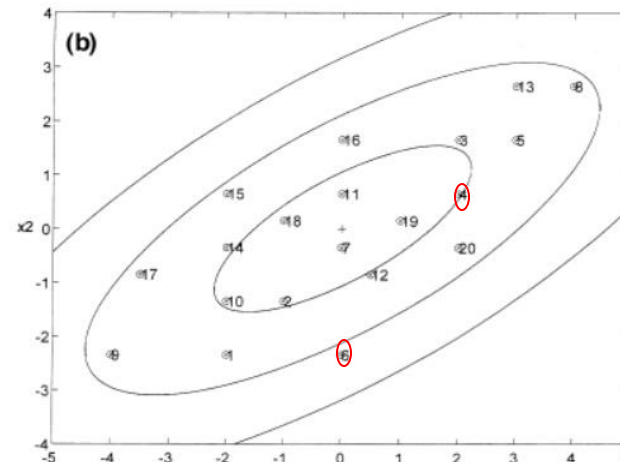
DATA DISTRIBUTION (2)

- The Mahalanobis distance
 - Let Σ be the covariance matrix

$$Maha(\bar{X}, \bar{Y}) = \sqrt{(\bar{X} - \bar{Y})\Sigma^{-1}(\bar{X} - \bar{Y})^T}.$$

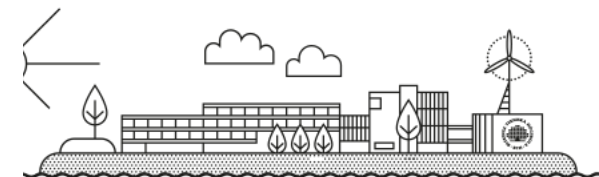


Points on each circle have the same Euclidean distance to the origin.



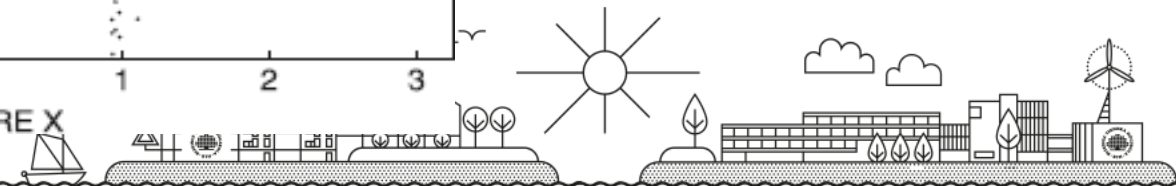
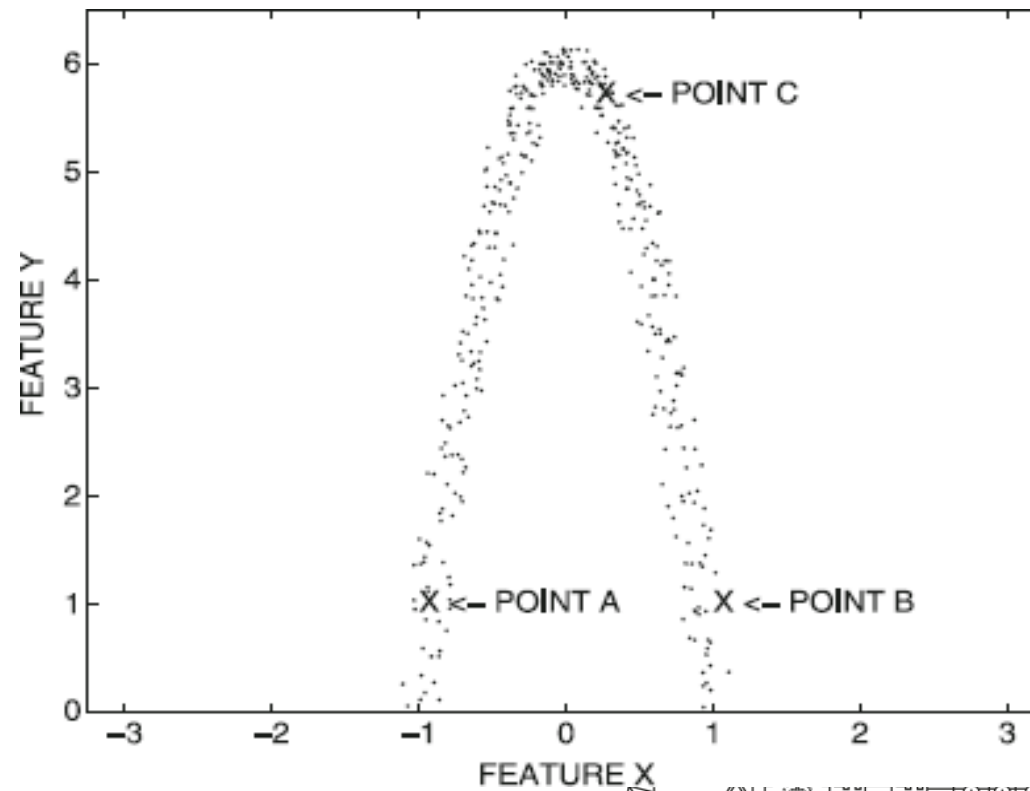
Points on each ellipsis have the same Mahalanobis distance to the origin.

<https://queirozf.com/entries/similarity-measures-and-distances-basic-reference-for-data-science-practitioners>



NONLINEAR DISTRIBUTIONS: ISOMAP (1)

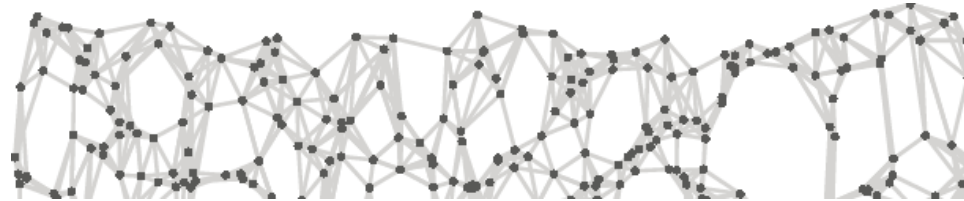
- Which one of B and C is closer to A?



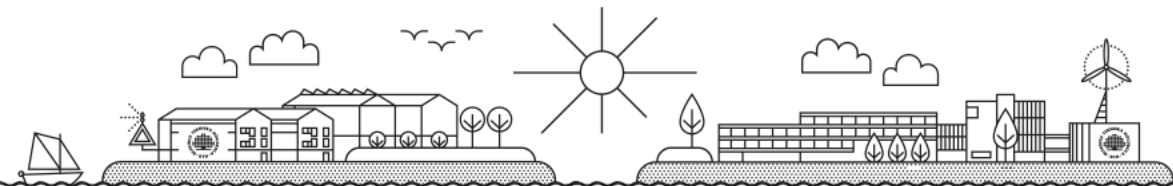
NONLINEAR DISTRIBUTIONS: ISOMAP (2)

○ Geodesic Distances

- Compute the k -nearest neighbors of each point
- Construct a weighted graph G with **nodes** representing data points, and **edge weights** representing (Euclidean) distance of these k -nearest neighbors

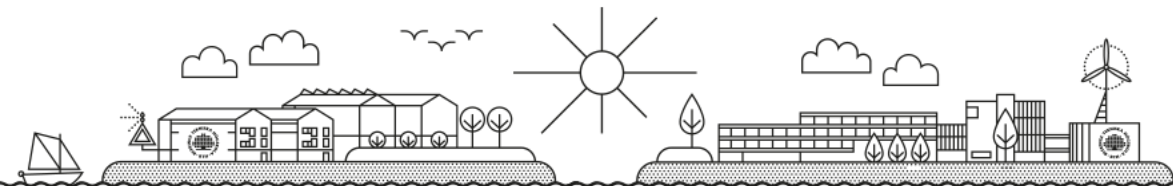


- $Dist(\bar{X}, \bar{Y})$ is the shortest path between any pair of points $\bar{X}$ and $\bar{Y}$ in the graph.



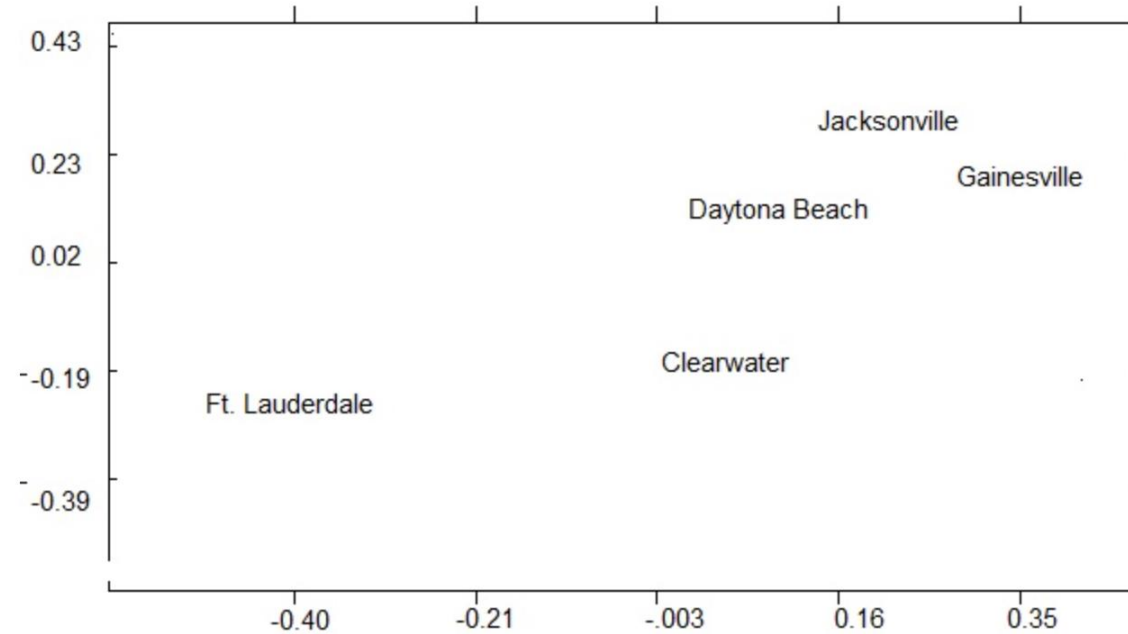
NONLINEAR DISTRIBUTIONS: ISOMAP (3)

- Nonlinear Dimensionality Reduction by ISOMAP
 - Compute the k -nearest neighbors
 - Construct a weighted graph G
 - Compute the distances between **all pairs** of data points
 - $A_{d \times d}$ distance matrix
 - Find vector representations by multidimensional scaling (MDS)
 - To embed distances into a 2-D map
 - $Dist(\bar{X}, \bar{Y})$ is the Euclidean distance of the new representations

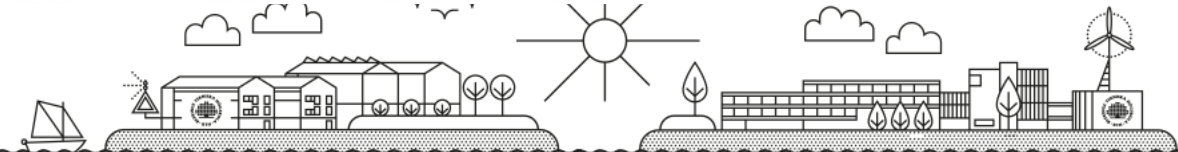


MULTIDIMENSIONAL SCALING (MDS)

CITY	Clearwater	Daytona Beach	Ft. Lauderdale	Gainesville	Jacksonville
Clearwater	0	159	247	131	197
Daytona Beach	159	0	230	97	89
Ft. Lauderdale	247	230	0	309	317
Gainesville	131	97	309	0	68
Jacksonville	197	89	317	68	0

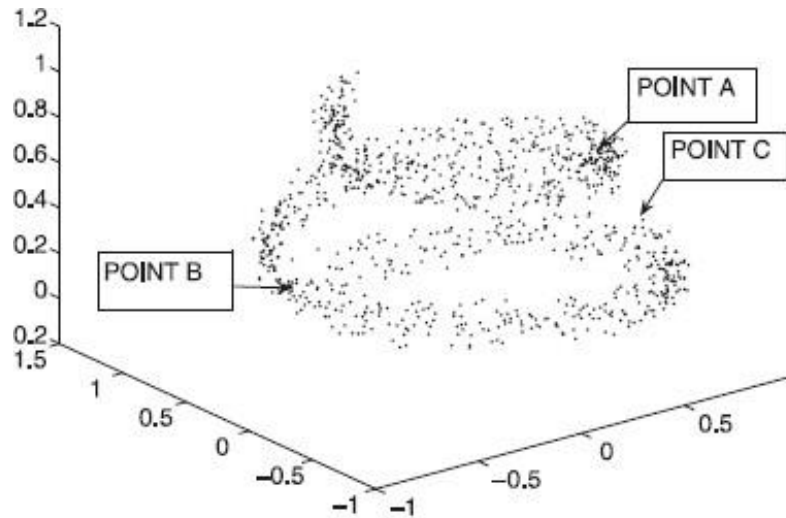


<https://www.statisticshowto.com/multidimensional-scaling/>

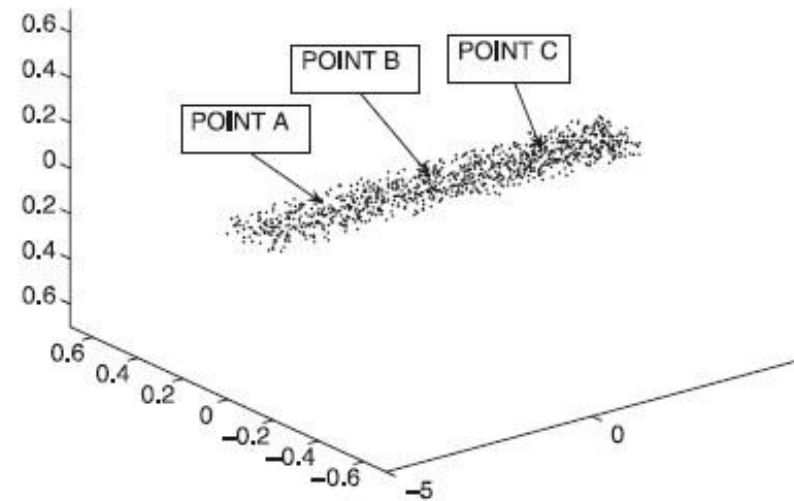


NONLINEAR DISTRIBUTIONS: ISOMAP (4)

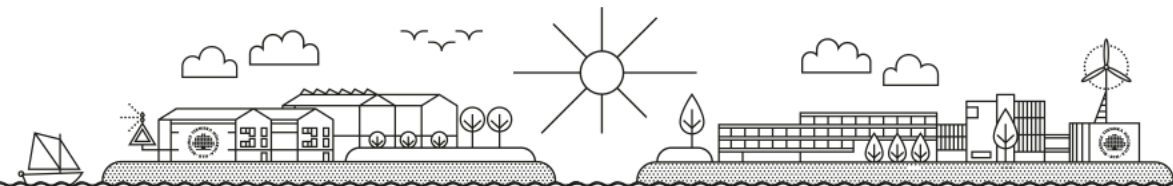
○ An Example of ISOMAP



(a) A and C seem close
(original data)



(b) A and C are actually far away
(*ISOMAP* embedding)



CATEGORICAL DATA (1)

○ Given $\overline{X} = (x_1 \dots x_d)$ and $\overline{Y} = (y_1 \dots y_d)$

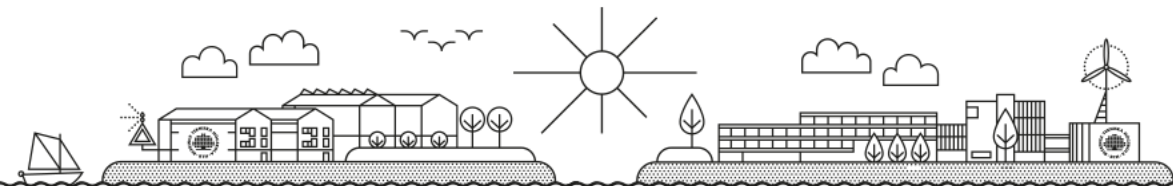
- Sum of similarities on the individual features

$$sim(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^k w_i s(x_i, y_i)$$

- Typically weight $w_i = \frac{1}{k}$ (k =number of features)
- Many choices for s , e.g., in overlap similarity s is

$$s(x_i, y_i) = \begin{cases} 1 & \text{if } x_i = y_i \\ 0 & \text{otherwise} \end{cases}$$

- Overlap similarity = fraction of dimensions where X and Y have an equal value



CATEGORICAL DATA (2)

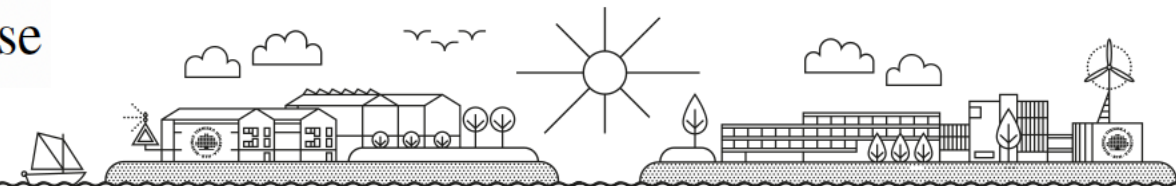
- E.g., a categorical attribute represents “Normal” for 99 % of the records, and either “Cancer” or “Diabetes” for the remaining.
- Take into account the frequency of value:

$$p_i(x_i) = \frac{fr(A_i = x_i)}{n}$$

fraction of records having $A_i = x_i$

- Goodall measure:

$$s(x_i, y_i) = \begin{cases} 1 - p_i^2(x_i) & \text{if } x_i = y_i \\ 0 & \text{otherwise} \end{cases}$$



SIMILARITY IN MIXED DATA

- Given $\bar{X} = (\bar{X}_n, \bar{X}_c)$ and $\bar{Y} = (\bar{Y}_n, \bar{Y}_c)$
 - Where $\bar{X}_n, \bar{Y}_n$ are the subsets of numerical attributes and $\bar{X}_c, \bar{Y}_c$ are the subsets of categorical attributes
 - Weighted Average

$$Sim(\bar{X}, \bar{Y}) = \lambda \cdot NumSim(\bar{X}_n, \bar{Y}_n) + (1 - \lambda) \cdot CatSim(\bar{X}_c, \bar{Y}_c)$$

- $\lambda = (0,1]$ domain specific otherwise can be the fraction of numerical features in data
- Normalized Weighted Average

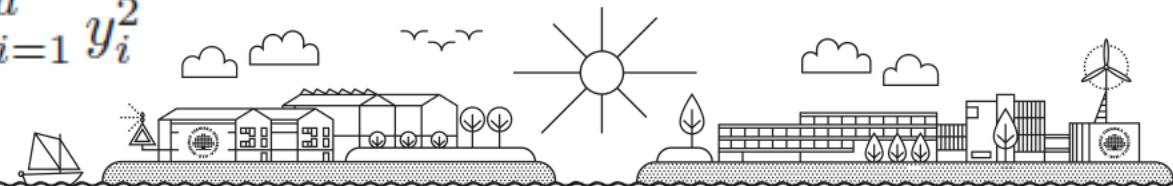
$$Sim(\bar{X}, \bar{Y}) = \lambda \cdot NumSim(\bar{X}_n, \bar{Y}_n) / \sigma_n + (1 - \lambda) \cdot CatSim(\bar{X}_c, \bar{Y}_c) / \sigma_c.$$



TEXT SIMILARITY MEASURES

- As Quantitative Multidimensional Data
 - Bag of Words model
 - It is very sparse
 - L_p -norm does not work well
 - Long documents have long distance
- Cosine Similarity
 - The angle between two documents

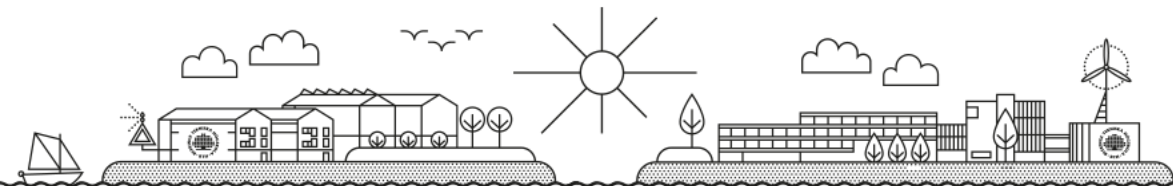
$$\cos(\bar{X}, \bar{Y}) = \frac{\sum_{i=1}^d x_i \cdot y_i}{\sqrt{\sum_{i=1}^d x_i^2} \cdot \sqrt{\sum_{i=1}^d y_i^2}}.$$



BINARY AND SET DATA

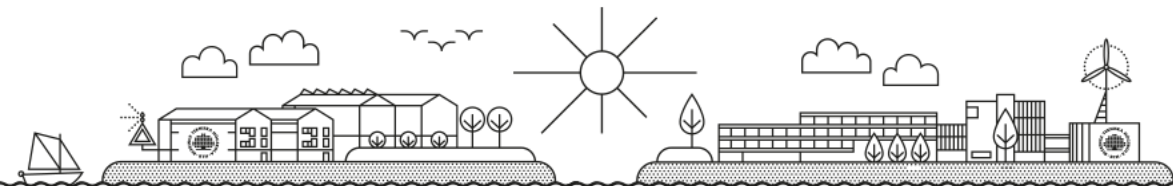
- Given $\bar{X} = (x_1 \dots x_d)$ and $\bar{Y} = (y_1 \dots y_d)$ with $x_i, y_i \in \{0,1\}$
 - They can be treated as vector representations of two sets
 - $S_X = \{i | x_i = 1\}$
 - $S_Y = \{i | y_i = 1\}$
 - Jaccard coefficient

$$J(\bar{X}, \bar{Y}) = \frac{\sum_{i=1}^d x_i \cdot y_i}{\sum_{i=1}^d x_i^2 + \sum_{i=1}^d y_i^2 - \sum_{i=1}^d x_i \cdot y_i} = \frac{|S_X \cap S_Y|}{|S_X \cup S_Y|}.$$

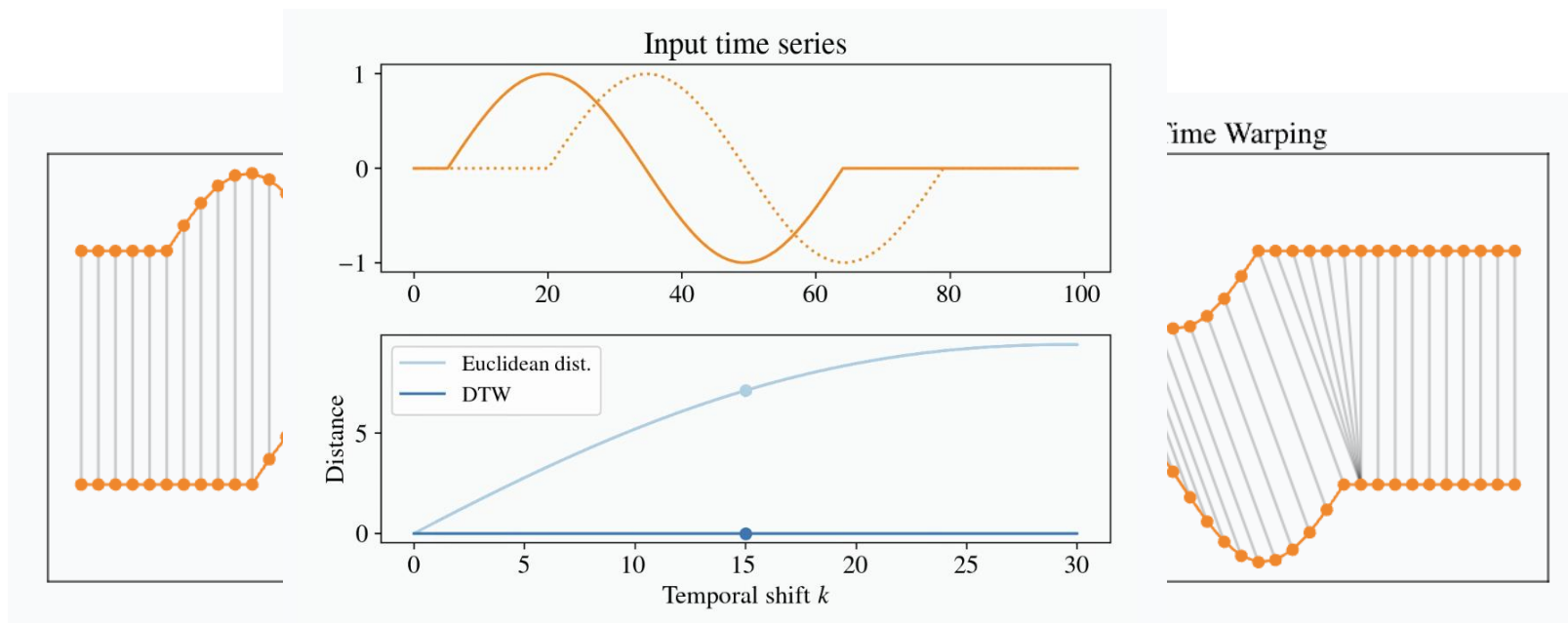


CONTINUOUS TIME SERIES

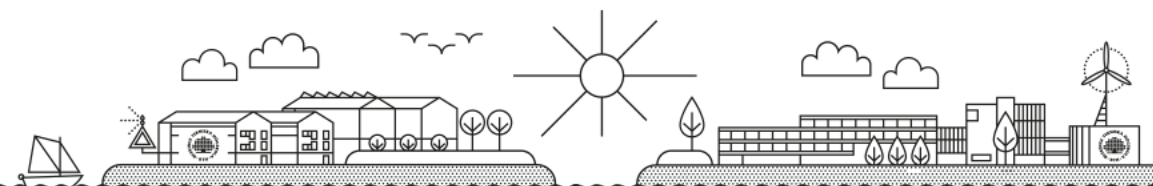
- If two time series $\bar{X}$ and $\bar{Y}$ are vectors of length d and corresponding dimensions represent the same point in time,
 - any vectorial distance function can be used to compare them
- Unfortunately, these assumptions are often violated in practice:
 - We often compare time series of different lengths, $d \neq \hat{d}$.
 - The time points at which the time series were observed are not synchronous.
 - The time intervals between observations may vary within and between time series.



CONTINUOUS TIME SERIES



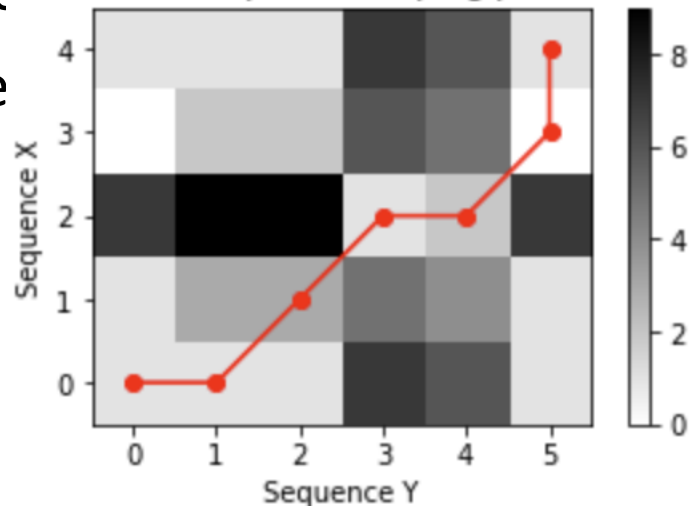
<https://rtavenar.github.io/blog/dtw.html>



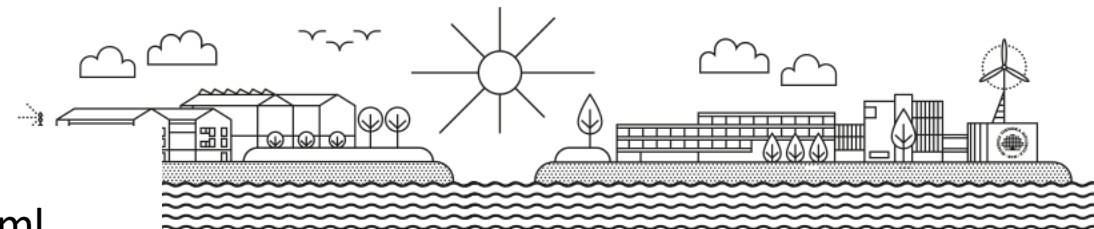
DYNAMIC TIME WARPING (DTW) (1)

- Given $\bar{X} = (x_1 \dots x_m)$ and $\bar{Y} = (y_1 \dots y_n)$
 - The two series have different lengths
- $DTW(i, j)$
 - The distance between the first i symbols of $\bar{X}$ and the first j symbols of $\bar{Y}$ may be based on Euclidean.

○ A Re

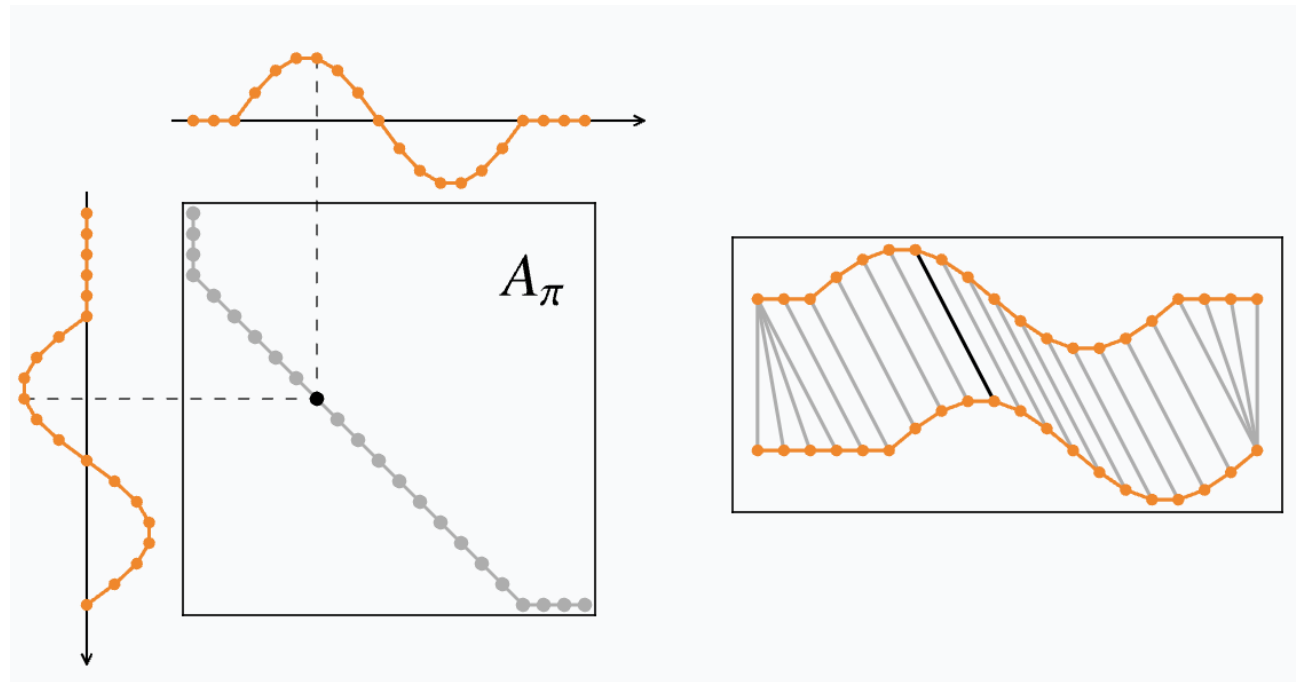


$$\min \begin{cases} DTW(i, j - 1) & \text{repeat } x_i \\ DTW(i - 1, j) & \text{repeat } y_j \\ DTW(i - 1, j - 1) & \text{repeat neither} \end{cases}$$

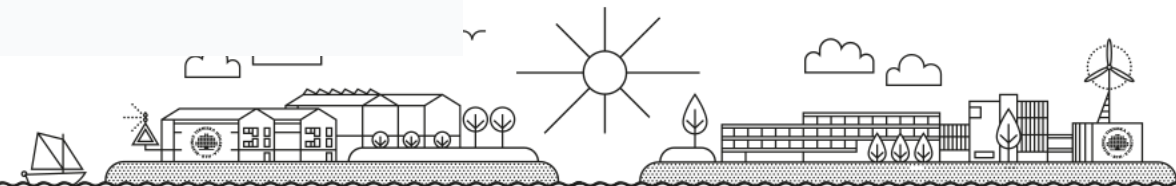


DYNAMIC TIME WARPING (DTW) (2)

- DTW is the cost of an optimal alignment between the measurements of two time series

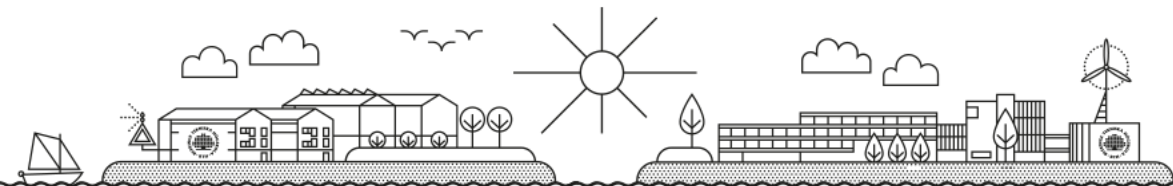


<https://rtavenar.github.io/blog/dtw.html>



EDIT DISTANCE (1)

- Edit Distance of Two Sequences
 - The cost of **edits** to transfer the first one to the second one
- Edits
 - Insertions
 - Deletions
 - Replacements
- Sequence **ababababab** to **bababababa**
 - Way 1: 10 Replacements
 - Way 2: 1 Deletion+1 Insertion



EDIT DISTANCE (2)

- Two Sequences $\bar{X} = (x_1 \dots x_m)$ and $\bar{Y} = (y_1 \dots y_n)$
 - $Edit(\bar{X}, \bar{Y})$ may not be the same as $Edit(\bar{Y}, \bar{X})$
- $Edit(i, j)$
 - The edit distance between the first i symbols of $\bar{X}$ and the first j symbols of $\bar{Y}$
- A Recursive Definition

$$Edit(i, j) = \min \begin{cases} Edit(i-1, j) + \text{Deletion Cost} \\ Edit(i, j-1) + \text{Insertion Cost} \\ Edit(i-1, j-1) + I_{ij} \cdot (\text{Replacement Cost}) \end{cases}$$

